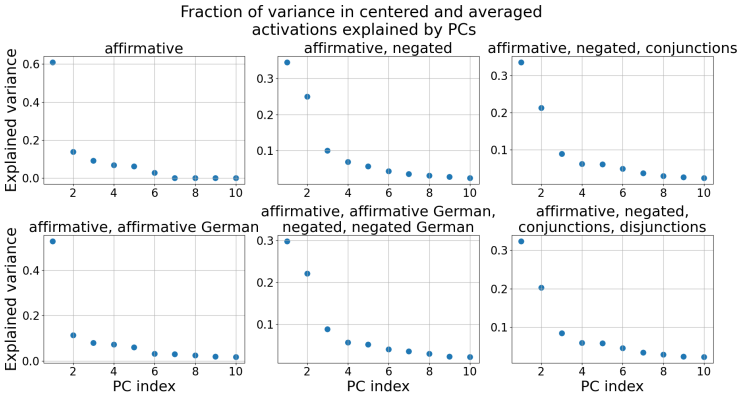


Figure 18: Mistral-7B: The fraction of variance in the centered and averaged activations $\tilde{\boldsymbol{\mu}}_i^+$, $\tilde{\boldsymbol{\mu}}_i^-$ explained by the Principal Components (PCs). Only the first 10 PCs are shown.

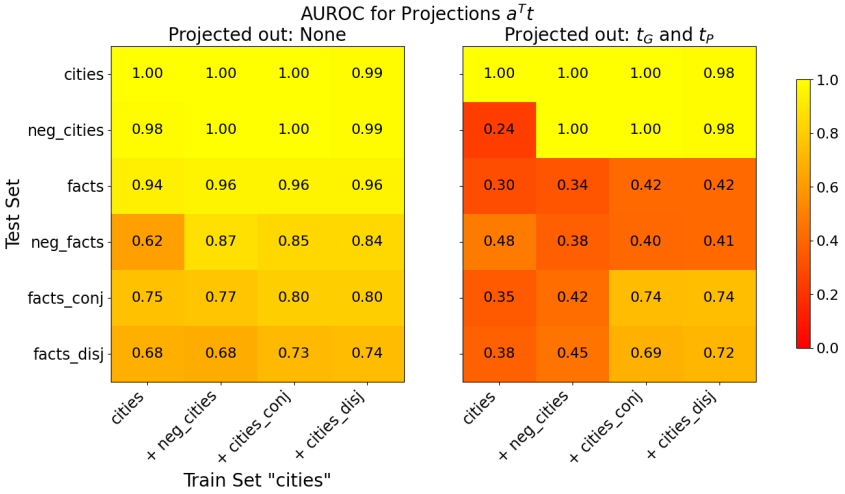
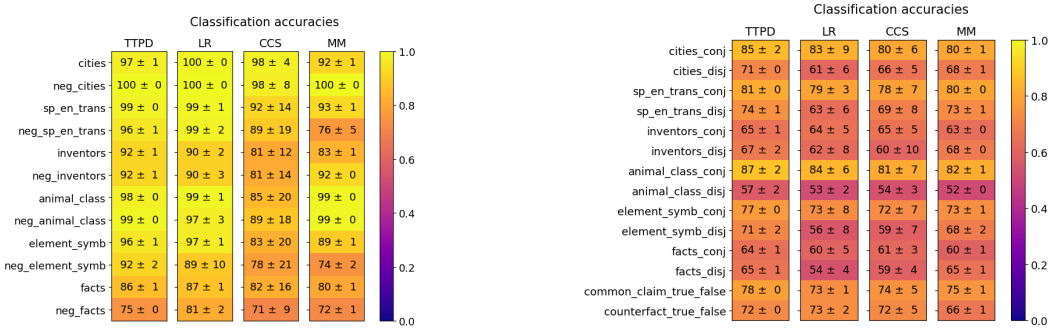


Figure 19: Mistral-7B: Generalisation accuracies of truth directions $\mathbf{t}$ before (left) and after (right) projecting out $\mathbf{t}_G$ and $\mathbf{t}_P$ from the training activations. The x-axis shows the train set and the y-axis the test set. All truth directions are trained on 80% of the data. If test and train set are the same, we evaluate on the held-out 20%, otherwise on the full test set. The displayed AUROC values are averaged over 10 training runs, each with a different train/test split.



(a) (b)

Figure 20: Mistral-7B: Generalization of TTPD, LR, CCS and MM. Mean and standard deviation are computed from 20 training runs, each on a different random sample of the training data.

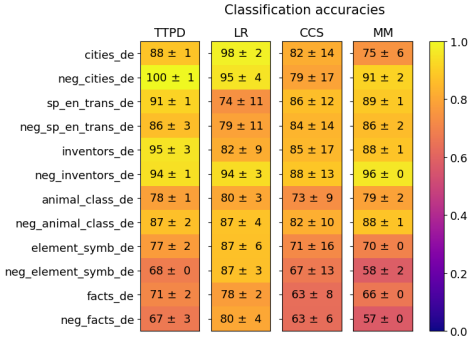


Figure 21: Mistral-7B: Generalization accuracies of TTPD, LR, CCS and MM on the German statements. Mean and standard deviation are computed from 20 training runs, each on a different random sample of the training data.

G.3 Gemma-7B

In this section, we present the results for the Gemma-7B-Instruct model. As shown in figure 22, the

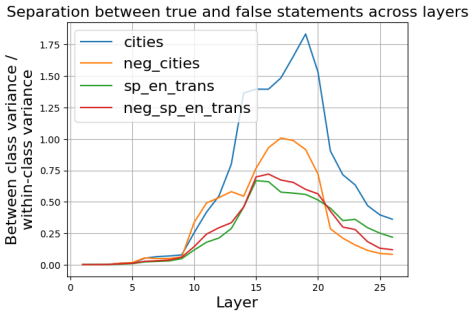
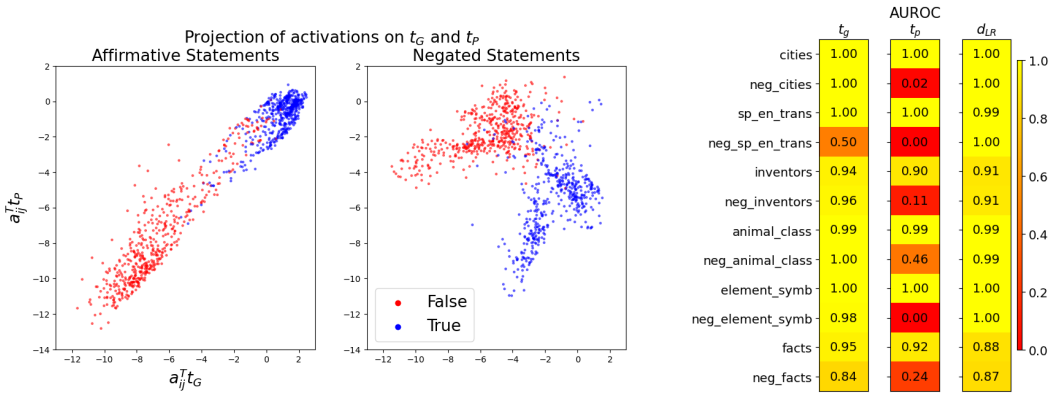


Figure 22: Gemma-7B: Ratio between the between-class variance and within-class variance of activations corresponding to true and false statements, across residual stream layers.

largest separation between true and false statements occurs in layer 16. Therefore, we use activations from layer 16 for the subsequent analysis of the Gemma-7B model. As can be seen in Figure 23, much higher classification accuracies would be possible by not only using $\mathbf{t}_G$ for classification but also $\mathbf{t}_P$.



(a) (b)

Figure 23: Gemma-7B: Left (a): Activations $\mathbf{a}_{ij}$ projected onto $\mathbf{t}_G$ and $\mathbf{t}_P$. Right (b): Separation of true and false statements along different truth directions as measured by the AUROC, averaged over 10 training runs.